

Written Assignment - Grammar Construction 1

Construct grammars for each language below.

1. $\{\omega \mid \omega \in \{a,b,c\}^* \text{ and the numbers of } a\text{'s and } b\text{'s and } c\text{'s occurred in } \omega \text{ are even}\}.$
2. $\{a^i b^j \mid i \geq (2j+1) \text{ and } j \geq 0\}$
3. $\{\omega \mid \omega \in \{a,b,c\}^* \text{ and the numbers of } a\text{'s and } b\text{'s occurred in } \omega \text{ are odd.}\}$
4. $\{\omega \mid \omega \in \{a,b,c\}^*, \omega \text{ is lead by a and the numbers of } a\text{'s and } b\text{'s occurred in } \omega \text{ are even } \}$
5. $\{\omega \mid \omega \in \{a,b,c\}^*, \text{ and } \omega \text{ starts with a and ends with b, the numbers of } a\text{'s and } c\text{'s occurred in } \omega \text{ are even}\}$
6. $\{a^{2m} \omega b^{2m} \mid m \geq 0, \omega \in \{c,d,e\}^* \text{ and } \omega \text{ does not contain the substring } cdd\}.$
7. Construct a grammar for binary numbers whose corresponding decimal number can be divided by 3.
8. For Boolean expressions in C language, please construct a regular grammar of their lexical patterns of at least 3 types of tokens, and a context-free grammar of sentential patterns containing at least 3 types of boolean operators.

Complete the homework and submit it before the next next class to the teaching assistant (Room 442/444, Building of Sch. Computer Sci&Eng).